PRDA – 05 CUSTOMER DATA ANALYSIS

INTERNSHIP CAPSTONE PROJECT

SUBMITTED BY:

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TOOL USED: MYSQL WORKBENCH, POWER BI

YEAR: 2026

TASK 1

How is the shopping distribution according to gender?

Objective

The objective of this analysis is to understand how shopping transactions are distributed between Male and Female customers.

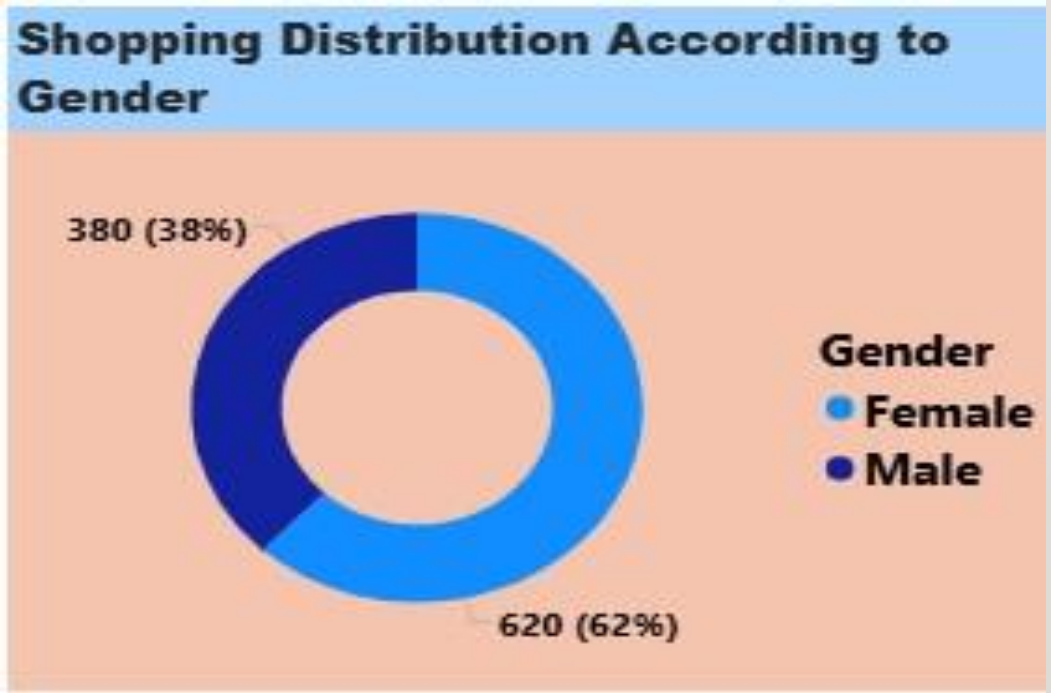
Description

A Donut Chart is used in Power BI to visualize the distribution of shopping transactions based on gender. This type of chart helps to clearly show the percentage contribution of each gender to the total number of transactions.

Steps to Follow

- Import the dataset into Power BI.
- Select the Donut Chart from the Visualizations pane.
- Drag the Gender field to the Legend section.
- Drag the Transaction Count / Customer ID to the Values section.
- Add the chart title "Shopping Distribution According to Gender."
- Enable Data Labels to display the values and percentages.

Visualization



Insights

From the chart, it is observed that Female customers contribute 62% of the total transactions, while Male customers contribute 38%. This indicates that Female customers make more purchases compared to Male customers.

TASK 2

Which gender did we sell more products to?

SQL Query:

```
SELECT gender,
 SUM(quantity) AS
Total_products_sold from
customer
Group BY gender;
```

Query 1

```
use project_customer_segmentation;
select gender, SUM(quantity) AS
total_products_sold
from customer
Group BY Gender;
```

Result Grid

gender	total_products_sold
Female	178659
Male	120053

Result 3

Read Only

Insights

Based on the total quantity sold, female customers purchased more products than male customers.

TASK 3

Which gender generated more revenue?

```
 Select gender,
Sum(quantity * price) As total_revenue
from customer
Group By gender;
```

Query 1

Limit to 1000 rows

```
select
gender,
sum(Quantity * price) As total_revenue
from customer
Group BY gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	total_revenue
Female	150207136.02
Male	101298658.23

Result Grid

Form Editor

Field Types

Result 5

Read Only

Insights

From the result, it is observed that female customers generated more revenue than male customers.

TASK 4&8

Distribution of Purchase Categories Relative to Gender?

To analyse how product categories are purchased relative to gender, two visualizations were created in Power BI.

Chart 1

Total Quantity Purchased by Category and Gender (Clustered Column Chart)

Steps to Follow

1.Import the dataset into Power BI.

2.Select the Clustered Bar Chart from the visualization pane.

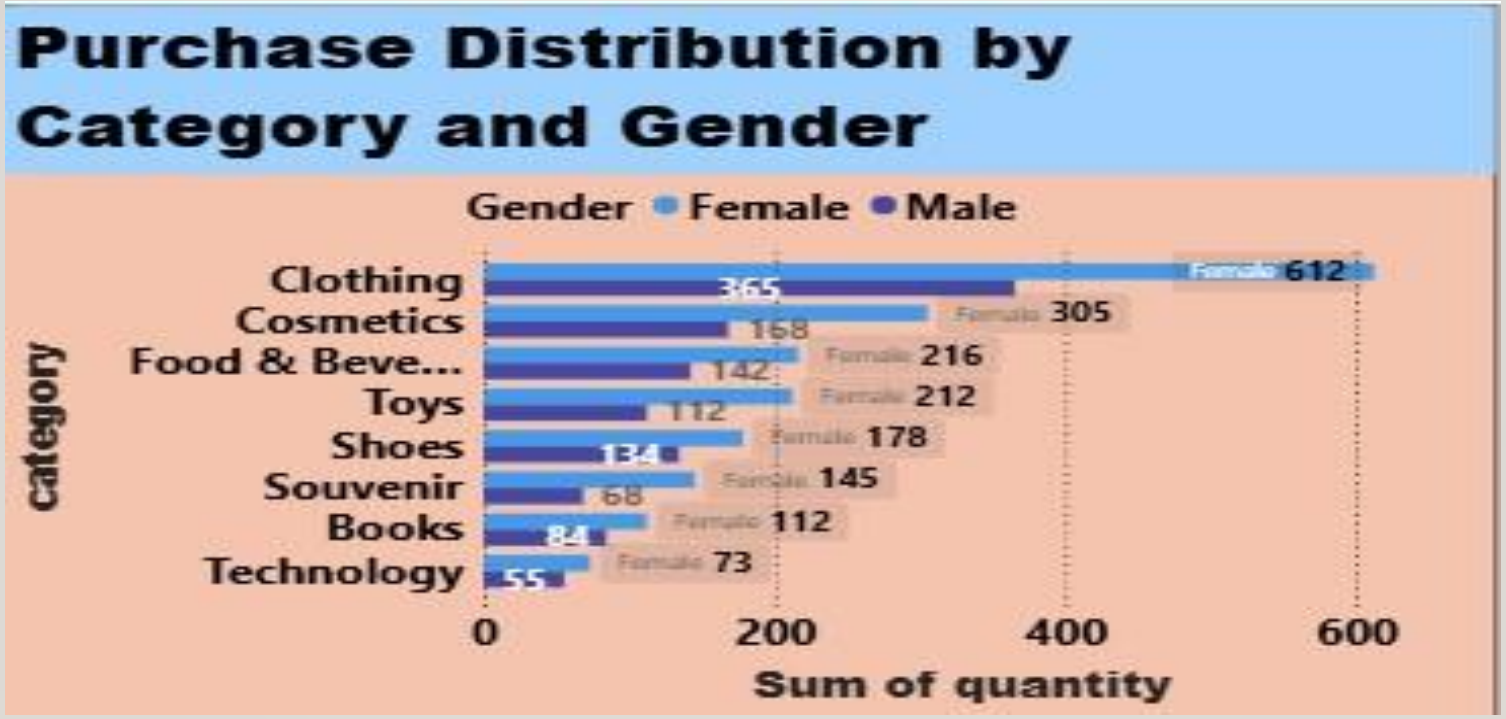
3. Drag the category field to the Y-axis.

4. Drag Quantity to the X-axis set it to Sum of Quantity.

5. Drag Gender to the Legend field to compare Male and Female purchases.

6. Add data labels and provide a suitable chart title.

Visualization



Description

This chart shows the total quantity of products purchased in each category and compares the purchases made by male and female customers.

Observation

Clothing has the highest quantity purchased among all categories. Female customers tend to purchase more items in categories like Clothing, Cosmetics, and Toys compared to male customers.

Insights

The chart highlights differences in purchasing patterns between male and female customers and helps identify which product categories are more popular among each gender.

Chart 2

Total Quantity Purchased by Category and Gender (Clustered Column Chart)

Steps to Follow

- Import the dataset into Power BI Desktop.
- From the Visualizations pane, select the Clustered Column Chart.
- Drag the Category field to the X-axis.
- Drag the Quantity field to the Y-axis and set it as Sum of Quantity.
- Drag the Gender field to the Legend section to compare Male and Female purchases.
- Enable Data Labels to clearly display the quantity values.
- Add a suitable title such as “Total Quantity Purchased by Category and Gender.”
- This chart shows a vertical comparison of total quantity purchased across different categories by male and female customers.

Visualization

Observation

The chart shows the total quantity of products purchased in each category by male and female customers. Clothing and Cosmetics categories have higher purchase quantities compared to other categories.

Insights

The visualization helps compare purchasing behaviour between male and female customers across different product categories and identifies which categories have higher purchase quantities

TASK 5

Shopping Distribution by Age Group

Objective

To analyze how shopping expenditure is distributed across different age groups.

Chart Used

Clustered Column Chart

- **<u>Steps to Follow</u>**
- Open Power BI Desktop.
- Import the dataset.
- Select Clustered Column Chart from the Visualizations pane.
- Drag Age Group to the X-axis.
- Drag Price to the Y-axis.
- Set Price aggregation to Sum.
- Turn Data Labels ON for better visibility.
- Add the title "Shopping Distribution by Age Group."

- **Observation**
- The chart shows the total shopping value for each age group. The 56+ age group has the highest total spending, while the 18–25 age group has the lowest spending.
- **Insights**
- Customers in the 56+ age group spend the most on shopping.
- The 36–45 age group shows moderate spending.
- The 18–25 age group contributes the least to total shopping value.

- Overall, shopping expenditure increases with age.

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Conclusion

The analysis indicates that older customers tend to spend more on shopping compared to younger customers.

TASK 6

- Which age category did we sell more products to?
- Step 1
- Find which age group bought more products
- Step 2
- Group customers by age and calculate total quantity

SQL Query

```
SELECT
  CASE
    WHEN age BETWEEN 0 AND 18 THEN 'Teen'
```

- WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
- WHEN age BETWEEN 31 AND 45 THEN 'Adult'
- WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
- ELSE 'Senior'
- END AS age_category,
- SUM(quantity) AS total_products_sold
- FROM customer
- GROUP BY age_category
- ORDER BY total_products_sold DESC;

Query 1

Limit to 1000 rows

Find

Done

```sql
SELECT
    CASE
        WHEN age BETWEEN 0 AND 18 THEN 'Teen'
        WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
        WHEN age BETWEEN 31 AND 45 THEN 'Adult'
        WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior'
    END AS age_category,
    SUM(quantity) AS total_products_sold
FROM customer
GROUP BY age_category
ORDER BY total_products_sold DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_category	total_products_sold
Adult	86604
Middle Age	85399
Young Adult	69654
Senior	51537
Teen	5518

Result 2 ×

Read Only

Insights

Based on the analysis the adult age group shows the highest product purchases, making it the most active customer segment.

TASK 7

Which age Category generated more revenue?

Step 1

Find which age group generated the highest revenue.

Step 2

Calculate total revenue using quantity and price and group customers by age category.

- **SQL Query**

```
SELECT
    CASE
      WHEN age BETWEEN 0 AND 18 THEN 'Teen'
      WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
       WHEN age BETWEEN 31 AND 45 THEN 'Adult'
       WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior’
  END AS age_category,
  SUM(quantity * price) AS total_revenue
FROM customer
GROUP BY age_category
ORDER BY total_revenue DESC;
```

Limit to 1000 rows

Find

```sql
SELECT
    CASE
        WHEN age BETWEEN 0 AND 18 THEN 'Teen'
        WHEN age BETWEEN 19 AND 30 THEN 'Young Adult'
        WHEN age BETWEEN 31 AND 45 THEN 'Adult'
        WHEN age BETWEEN 46 AND 60 THEN 'Middle Age'
        ELSE 'Senior'
    END AS age_category,
    SUM(quantity * price) AS total_revenue
FROM customer
GROUP BY age_category
ORDER BY total_revenue DESC;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_category	total_revenue
Adult	73134764.61
Middle Age	71804157.87
Young Adult	58649681.85
Senior	43519469.66
Teen	4397720.26

Result 3 × Read Only

TASK 9

- Does the payment method have a relation with other columns?

Objective

- The objective of this analysis is to examine the total revenue generated through different payment methods and genders and to understand the shopping distribution across various age groups.

Chart 1

- Total Revenue by Payment Method and Gender

Steps to Follow

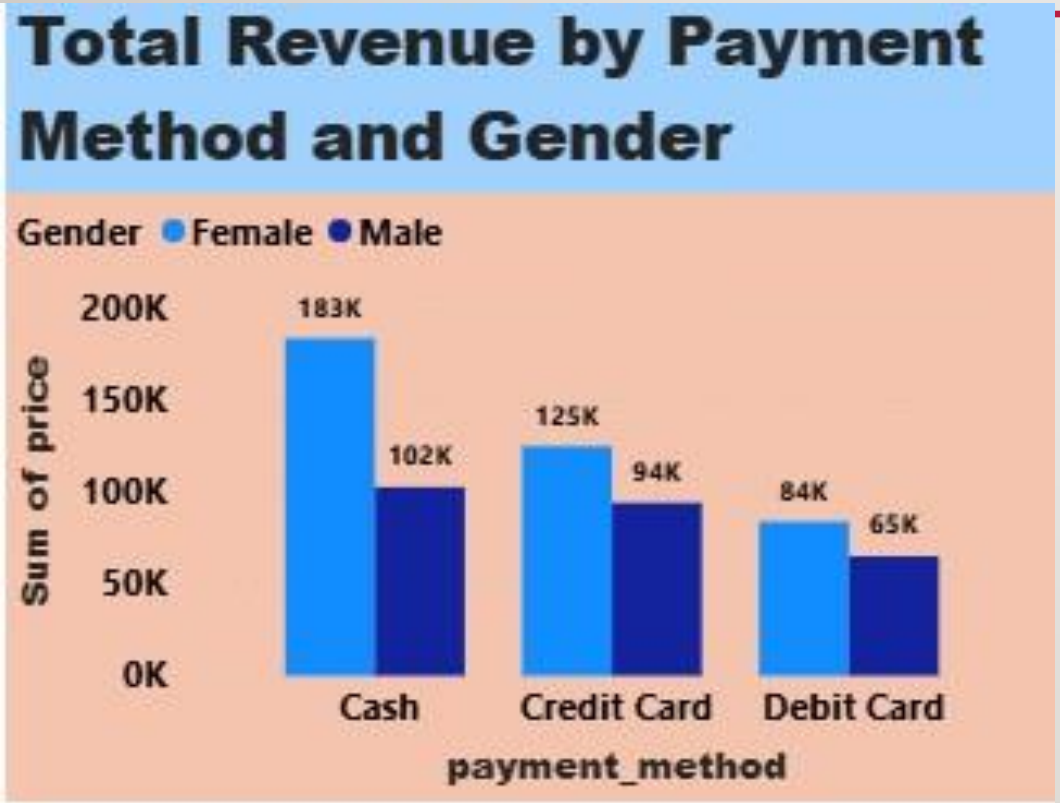
- Import the dataset into Power BI.
- Select Clustered Column Chart from the Visualizations panel.
- Drag payment method to the X-axis.
- Drag price to the Y-axis to calculate Sum of price (Total Revenue).
- Drag gender to the Legend section.
- Enable Data Labels for better visibility.
- Add the chart title "Total Revenue by Payment Method and Gender."

Description

- This chart shows the total revenue generated through different payment methods such as Cash, Credit Card, and Debit Card. The data is further categorized by gender to compare the revenue contribution of male and female customers.

Insights

- Female customers generate higher revenue compared to male customers across most payment methods. Cash is the most preferred payment method and contributes the highest revenue.


Total Revenue by Payment Method and Gender
Gender Female Male
Sum of price
200K
150K
100K
50K
0K
183K
102K
125K
94K
84K
65K
Cash
Credit Card
Debit Card
payment_method

Shopping Distribution by Age Group
Sum of price
200K
150K
100K
50K
0K
100K
108K
131K
108K
206K
18-25
26-35
36-45
46-55
56+
Age Group

Chart 2

- Shopping Distribution by Age Group

Steps to Follow

- Insert another Clustered Column Chart.
- Drag Age Group to the X-axis.
- Drag price to the Y-axis to calculate Sum of price.
- Sort the Age Group in ascending order.
- Enable Data Labels to show the values clearly.
- Add the chart title “Shopping Distribution by Age Group.”

Description

- This chart illustrates the total shopping revenue generated by different age groups such as 18–25, 26–35, 36–45, 46–55, and 56+.

Insights

The chart indicates that the 56+ age group contributes the highest shopping revenue, while the 18–25 age group contributes the lowest revenue.

TASK 10

How is the distribution of the payment method?

Objective

- To analyze how customers prefer to pay for their purchases and understand the proportion of different payment methods used in transactions.

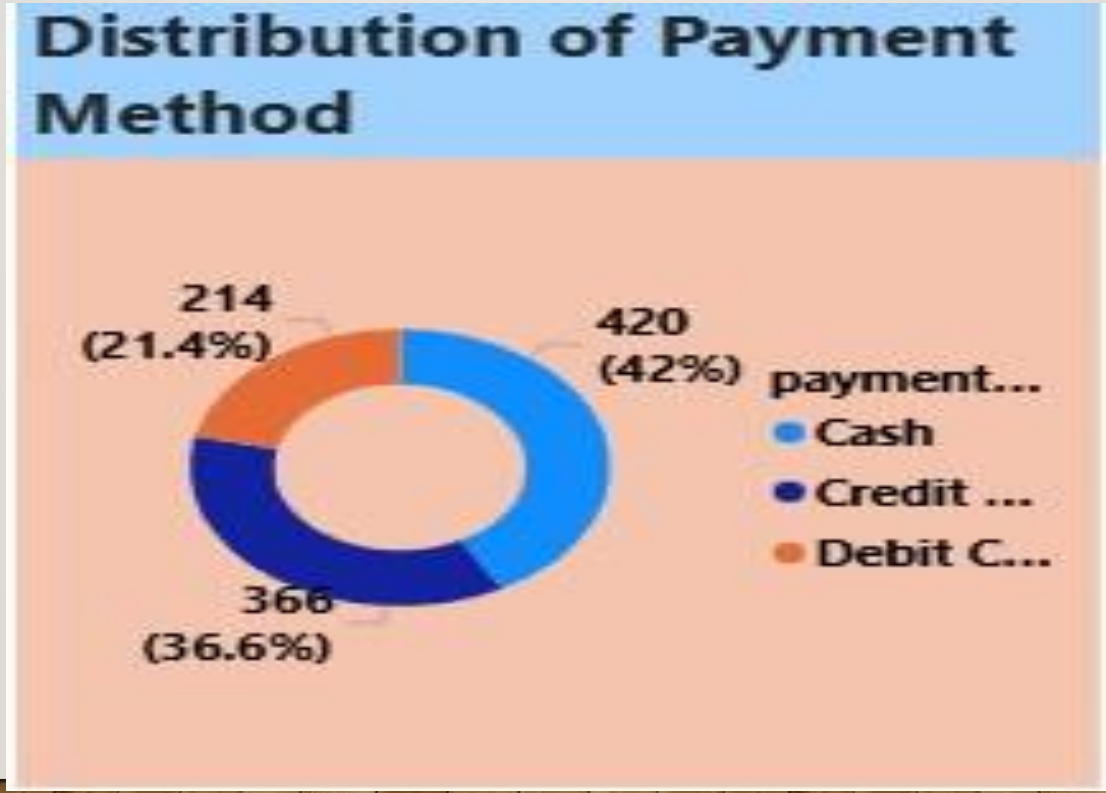
Chart Used

- Donut Chart
- A donut chart is used because it clearly shows the percentage distribution of payment methods.

Steps to Create in Power BI

- Open Power BI Desktop.
- Go to Visualizations panel.
- Select Donut Chart.
- Drag payment method → Legend.
- Drag invoice no → Values.
- Change aggregation to Count of invoice no.
- Turn Data Labels → ON.
- Turn Percentage → ON.
- Add chart Title: "Distribution of Payment Method".

Visualizations



Insights

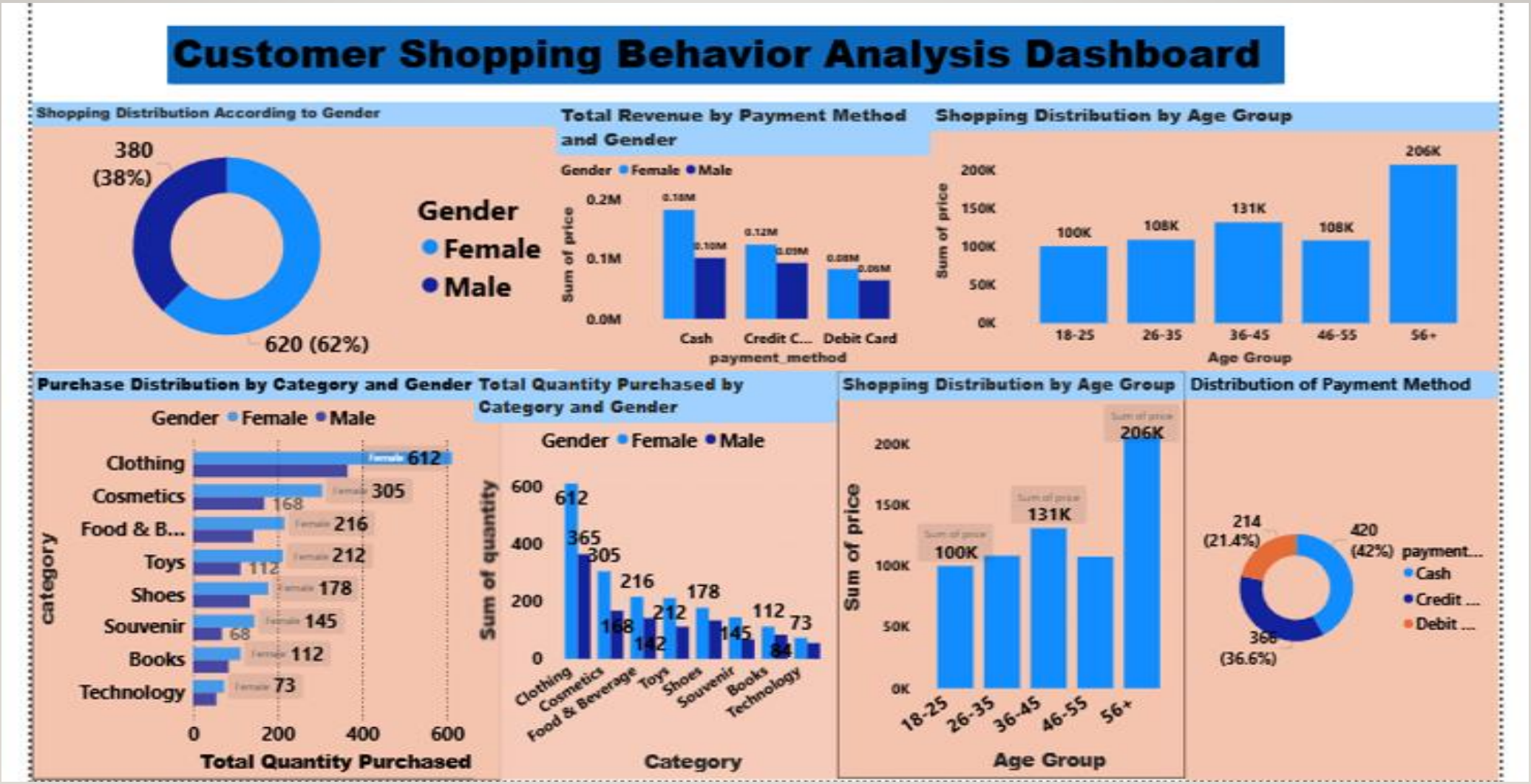
The chart shows the distribution of different payment methods used by customers. Cash payments represent the largest share of transactions at 42%, followed by Credit Card payments at 36.6%, while Debit Card payments account for 21.4%. This indicates that most customers prefer paying with cash compared to digital payment methods.

Conclusion

Cash is the most commonly used payment method, whereas debit card transactions are relatively lower. Businesses may encourage digital payments to improve transaction efficiency and tracking.

TASK 11

Visualize the data using Tableau /Power BI and derive insights and give your inputs/suggestions to the company


Customer Shopping Behavior Analysis Dashboard
Shopping Distribution According to Gender
380 (38%)
620 (62%)
Gender
Female
Male
Total Revenue by Payment Method and Gender
Gender Female Male
Sum of price
0.2M
0.1M
0.0M
0.10M
0.12M
0.09M
0.08M
0.06M
Cash
Credit C...
Debit Card
payment_method
Shopping Distribution by Age Group
Sum of price
200K
150K
100K
50K
0K
100K
108K
131K
108K
206K
18-25
26-35
36-45
46-55
56+
Age Group
Purchase Distribution by Category and Gender
Gender Female Male
category
Clothing
Cosmetics
Food & B...
Toys
Shoes
Souvenir
Books
Technology
612
305
168
216
212
112
178
145
68
112
73
0
200
400
600
Total Quantity Purchased
Total Quantity Purchased by Category and Gender
Gender Female Male
Sum of quantity
600
400
200
0
612
365
305
168
216
142
212
178
145
112
84
73
Clothing
Cosmetics
Food & Beverage
Toys
Shoes
Souvenir
Books
Technology
Category
Shopping Distribution by Age Group
Sum of price
200K
150K
100K
50K
0K
100K
131K
206K
18-25
26-35
36-45
46-55
56+
Age Group
Distribution of Payment Method
214 (21.4%)
420 (42%)
366 (36.6%)
payment...
Cash
Credit ...
Debit ...

Conclusion

This project used Microsoft Power BI to analyse customer shopping behaviour and visualize key insights from the dataset. The dashboard highlights patterns in purchasing based on gender, age group, product category, and payment methods. These visual insights help the company better understand customer preferences and improve business decision-making strategies. Overall, the dashboard provides a clear and effective way to interpret the data and support data-driven decisions.